

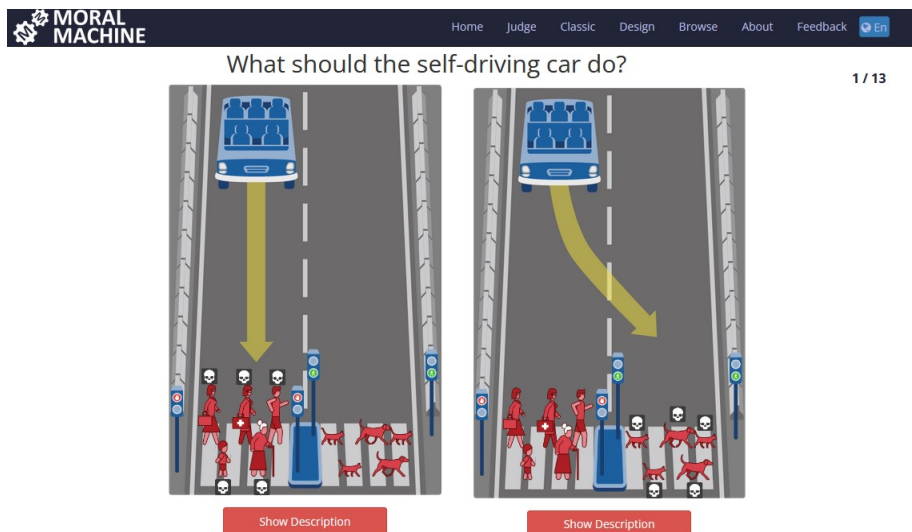
Exercises: Decision-Making

Goals

1. Understand Utility Theory
2. Apply Utility Theory and MU criteria
3. Apply Utility Theory and MEU criteria
4. Apply Utility Theory and Bayes Inference
5. Apply Utility Theory and MEU criteria in a game theory problem

Exercises

1. Moral dilemmas for Humans and AIs.



- Go to the “Moral Machine experiment” page at MIT:
<https://www.moralmachine.net/>
 - Start judging the scenarios.
 - As a Human: ¿how did you make the decisions?¿Which criteria have you followed?
2. Make a plot where the X axis represents “Amount of Money” between 0 and 10 million euros, and the Y axis is the utility of Money:
 - ¿How do you represent the utility function for you?
 - Consider the case you have 0 euros in your account: ¿Would it change your utility function?
 - In the case of having 1 million euros in your account: ¿how would the function be?
 - Compare your plots with the theory of marginal utility proposed by Bernoulli. In this link you have a comprehensive description of his ideas:

3. You are an AI expert of an important business company in the touristic sector. The firm manages a business of creation of touristic packages that are sold to travel agencies. The goal of the firm is to build a recommender system of packages to generate recommendations for tourist. The packages are characterized by the following attributes:
- Quality of package (Number of stars of the hotel): 2,3,4, y 5.
 - Price: Valores between 120 and 870 euros.
 - Duration of the package: 5 or 7 days.
 - Visual appeal: High or Low

A field experiment was carried out with a sample of potential clients. 8 choice situations with 3 packages/alternatives were shown to each subject. The subject had to choose an alternative for each situation. The results were collected in the dataset "dataset_eleccion_hotel.xls". This is your task:

- Binarize the attributes of the packages
- Estimate the preference vectors for subjects 1 and 4.
- Make a search in Booking for hotels in Santiago. Select two hotels. Thereafter, on the basis of the preference vectors estimated in the previous point, estimate the utilities of the hotels for subjects 1 and 4. Which hotel would they choose?

4. Choose your favorite option:

Game 1

Option A: 2.500 euros with probability 0.33 2.400 euros with probability 0.66 0 with probability 0.1	Option B: 2.400 euros for sure.
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Game 2

Option A: 2.500 euros with probability 0.33 0 euros with probability 0.67	Option B: 2.400 euros with probability 0.34 0 with probability 0.66
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Game 3

Option A: 4.000 euros with probability 0.80 0 with probability 0.2	Option B: 3.000 euros for sure.
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Game 4

Option A: Tour of 3 weeks in Europe with probability 0.5	Option B: Tour of 1 week in Europe for sure.
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Game 5

Option A: 6.000 euros with probability 0.45	Option B: 3.000 euros with probability 0.9.
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- Estimate the utility using MEU. Match your choices with the ones predicted by MEU: ¿Are they the same? ¿Why?
 - Read the Kahneman/Tversky paper where they explain why MEU does not always work. The games presented above are of the same type of the games used in their paper. Check if your choices are similar to the majority of subjects that participate in the study.
5. The Umbrella Problem. You have to make a decision for tomorrow: ¿should you put the umbrella in your backpack or leave it at home? Make the decision under the following conditions:
- ¿Ask a senior local person for help! On the basis of her prediction, ¿what will be your decision?
 - Your forecast app is working and you have time to make some detailed calculations. Check the forecast for tomorrow and take a note of the available probabilities. ¿Will you put your umbrella in your pack or leave it at home?
6. The Umbrella-App Problem. You have the following data:
- Prior information: your own experience.
 - Your forecast app is not so accurate in the last few weeks, especially when its rainy. You have annotated its accuracy and you come up with these figures:
 - i. $P(F=\text{Sun}|W=\text{Sun})= 0,6$
 - ii. $P(F=\text{Rain}|W=\text{Sun})= 0,4$
 - iii. $P(F=\text{Sun}|W=\text{Rain})= 0,55$
 - iv. $P(F=\text{Rain}|W=\text{Rain})= 0,45$
- Answer these questions:
- ¿What will be your decision if the app says that tomorrow is going to rain?
 - ¿Would you choose a different alternative if you focus only on your prior information?
7. One of your friends sells a book entitled “Python”. It is a brand new edition, but your friend is not interested on it and published an offer of just 10 euros in Wallapop. As you are really interested in “Python programming” (aren’t you? ;-)), you talk with your friend and he tells you he does not know whether the topic is “Programming/Language” or “Animals/Snake”. He just remembers having read some words from the document:

Words
Dynamic
Long

In order to decide if you buy the book or not, you need to make some calculations to estimate the topic of the book.

First, you look for a database of books containing examples of both topics. You find that a classification can be made on the basis of the words or features found in the book:

Features	Classification
Pythons are constrictors that feed on birds and mammals	Snake
Python was originally developed as a scripting language	Language
A 49-ft.-long python was found in Indonesia	Snake
Python has a dynamic type system	Language
Python with vivid scales	Snake
Open source project	Language

The database has a downloadable dataset offering the probability that a book may be classified as “Programming Language” or “Animals/Snake” depending on the words found in the book:

Palabra/Atributo	Language	Snake
Dynamic	0.6	0.1
Constrictor	0.0	0.6
Long	0.1	0.2
Source	0.3	0.1
And	0.95	0.95

Second, you recall on the tastes of your friend and you estimate that the chance of having a book about “Programming/Language” is very low, about 20%.

With this information in your hand, estimate the utilities for all the possible outcomes, and make a decision if you buy the book or not.

* Note: If we assume the independence of the attributes that characterize the books, then the probability $P(D|C_i)$ can be calculated as follows:

$$P(D|C_i) = \prod_j P(A_j|C_i)$$

8. Prisoner's dilemma. The problem is stated as follows:

You and your partner belong to a criminal gang. The police have found some evidences against you and have decided to arrest you both. Each prisoner is in solitary confinement with no means of communication with their partner. The principal charge would lead to a sentence of 10 years in prison; however, the police do not have a solid evidence for a conviction. They plan to sentence both to 2 years in prison on a lesser charge, but offer each prisoner a [Faustian bargain](#): If one of them confesses to the crime of the principal charge, betraying the other, they will be pardoned and free to leave while the other must serve the entirety of the sentence (10 years) instead of just 2 years for the lesser charge. If both of you confesses, the principal charge of 10 years will be applied to both of you.

You have to make a decision: “confess” or “remain silent”:

- Create a table with the utility values per each case.
- Your partner is really a good friend, so you trust he will make the best for both of you.
- ¿What do you decide?: “confess” or “remain silent”?
- Now imagine the case that your partner is a ratbag, so you do not trust him. How does this fact modify your decision?

9. Revisiting the moral dilemmas for Humans and AIs.

- Recall what you learnt about Rational DM and Utility Theory.
- Now, ¿How would you program the AI of the self-driving car to make the decisions shown in the “Moral Machine experiment”?

Deliverables:

- **Exercises:**
 - **Session 1:** 6 and 7.
 - **Session 2:** 2 and 4 (only the part of the Sprinkler problem).
- **Delivery method:** Virtual Campus
- **Deadline:** Sunday, 6th April